Experiment 20

Experiment to understand K-means clustering algorithm for a given data set.

Aim:

Understand the K-means clustering algorithm for the dataset given.

Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df=pd.read_csv('Mall_Customers.csv')

df.info()

df.head()

sns.pairplot(df)

features=df.iloc[:,[3,4]].values

from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
```

```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()

sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
   .map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)") \
   .add_legend();
plt.show()

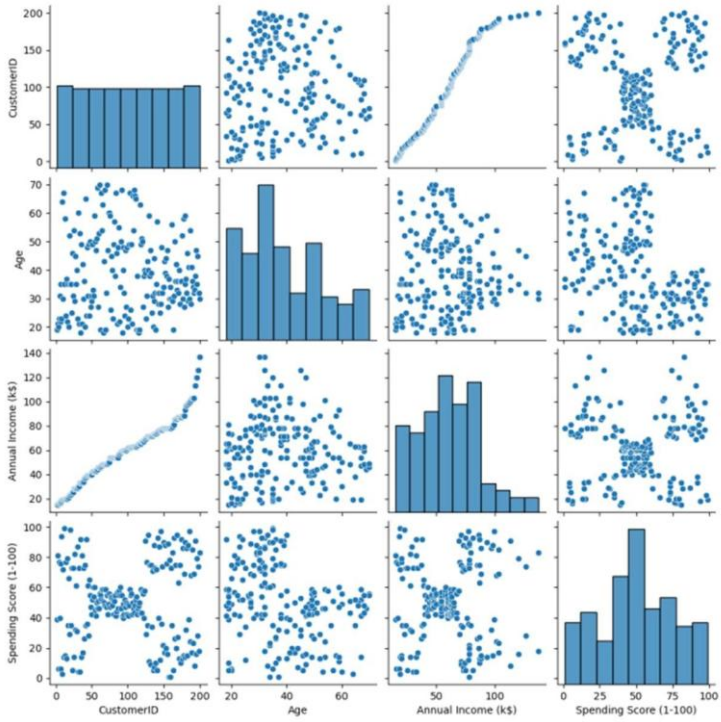
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Output:

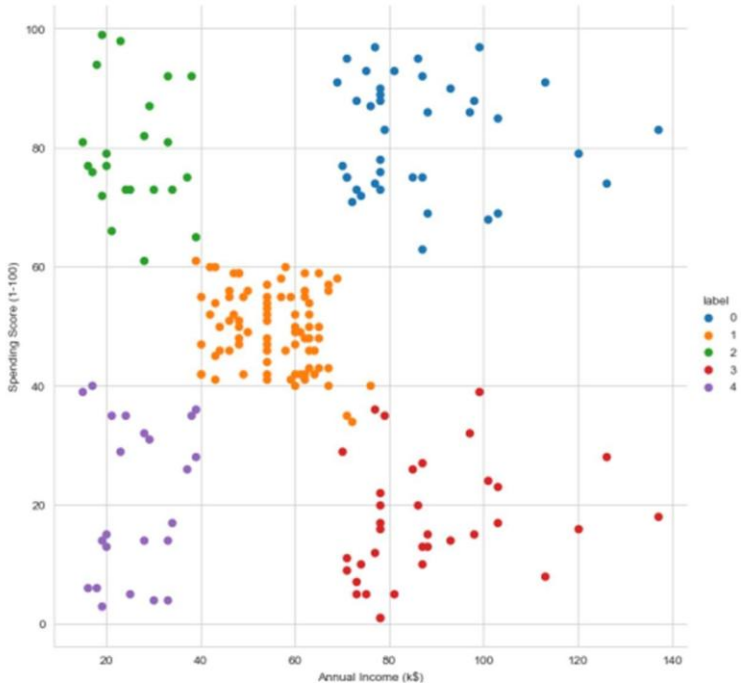
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

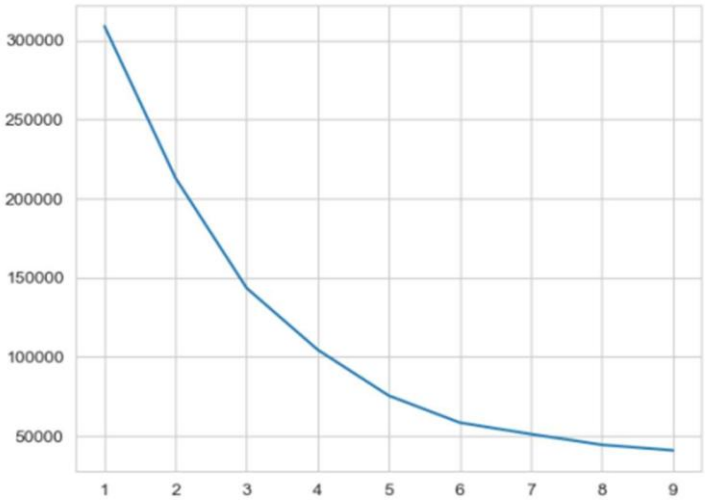
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
<seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4





Result:

K-Means clustering on the Mall Customers dataset successfully identified 5 distinct customer segments based on Annual Income and Spending Score. The Elbow Method confirmed k=5 as the optimal cluster count. Scatter plots clearly visualized the clusters, enabling targeted marketing strategy recommendations.